

1 LLM Model Comparison

To evaluate the effectiveness of the proposed Graph-RAG system, three different Large Language Models (LLMs) were tested using the same knowledge graph context and user queries. The comparison focuses on both quantitative and qualitative aspects, including response time, cost, correctness, and answer quality.

1.1 Evaluated Models

The following models were evaluated:

- **FLAN-T5-base** (open-source, HuggingFace)
- **FLAN-T5-large** (open-source, HuggingFace)
- **GPT-3.5-Turbo** (OpenAI API)

1.2 Test Queries

All models were evaluated using the same set of test queries to ensure fairness:

1. Which flights from Cairo to Dubai have high delays?
2. Which routes from CAI to DXB have delays more than 30 minutes?
3. Are there any severely delayed flights between Cairo and Dubai?

1.3 Quantitative Evaluation

The quantitative evaluation compares approximate response time, token usage, and cost. Token usage is not billed for open-source models.

Table 1: Quantitative Comparison of LLMs

Model	Avg. Response Time	Token Usage	Cost	Notes
FLAN-T5-base	~0.4 s	N/A	Free	Fast, lightweight
FLAN-T5-large	~1.1 s	N/A	Free	Higher accuracy
GPT-3.5-Turbo	~1–2 s	400–600	Paid	Best reasoning quality

1.4 Qualitative Evaluation

A qualitative, human-based evaluation was conducted to assess answer correctness, relevance, naturalness, and reasoning quality.

Table 2: Qualitative Comparison of LLMs

Model	Correctness	Relevance	Naturalness	Reasoning	Overall
FLAN-T5-base	Medium	Medium	Low	Low	Fair
FLAN-T5-large	High	High	Medium	Medium	Good
GPT-3.5-Turbo	High	High	High	High	Excellent